ASSIGNMENT 8

Basic SQL Queries and Aggregate Functions

Online Shopping Management System

1. Aim

To retrieve and summarize online shopping data using filtering, sorting, grouping, HAVING, and aggregate functions.

2. WHERE — filter records

```
SELECT ProductName, Category, Price
FROM Products
WHERE Price > 1000
ORDER BY Price DESC;
```

ProductName	Category	Price
Laptop	Electronics	55000.00
Smartphone	Electronics	25000.00
Office Chair	Furniture	4500.00
Headphones	Accessories	1500.00

Explanation: Returns products priced above 1,000, sorted from highest to lowest price.

3. ORDER BY — sort records

```
SELECT ProductName, Stock FROM Products ORDER BY Stock ASC;
```

ProductName	Stock
Office Chair	8
Laptop	10
Smartphone	20
Keyboard	25
Headphones	30

Explanation: Lists products from lowest stock to highest stock.

4. GROUP BY — count products per category

```
SELECT Category, COUNT(*) AS ProductCount
FROM Products
GROUP BY Category
ORDER BY Category;
```

Category	ProductCount
Accessories	2
Electronics	2
Furniture	1

Explanation: Groups products by category and counts the records in each group.

5. HAVING — filter grouped results

```
SELECT Category, COUNT(*) AS ProductCount
FROM Products
GROUP BY Category
HAVING COUNT(*) >= 2
ORDER BY Category;
```

Category	ProductCount
Accessories	2
Electronics	2

Explanation: HAVING filters groups after aggregation; only categories with at least two products remain.

6. Aggregate functions

COUNT()

```
SELECT COUNT(*) AS TotalCustomers FROM Customers;
```

TotalCustomers
5

Counts customer records.

SUM()

```
SELECT SUM(Stock) AS TotalUnitsInStock FROM Products;
```

TotalUnitsInStock
93

Adds the stock quantities across products.

AVG()

```
SELECT ROUND(AVG(Price), 2) AS AverageProductPrice FROM Products;
```

AverageProductPrice
17360.00

Calculates the average listed product price.

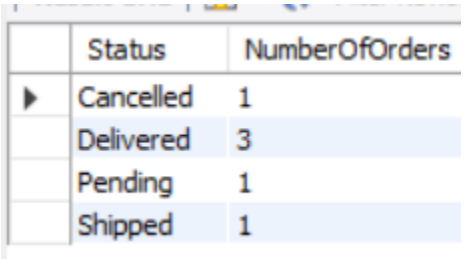
MIN() and MAX()

```
SELECT MIN(Price) AS LowestPrice, MAX(Price) AS HighestPrice FROM Products;
```

LowestPrice	HighestPrice
800.00	55000.00

7. Order status summary

```
SELECT Status, COUNT(*) AS NumberOfOrders
FROM Orders
GROUP BY Status
ORDER BY Status;
```



Status	NumberOfOrders
Cancelled	1
Delivered	3
Pending	1
Shipped	1

8. Conclusion and evidence

The queries demonstrate WHERE, ORDER BY, GROUP BY, HAVING, COUNT(), SUM(), AVG(), MIN(), and MAX(). Outputs shown are expected for the sample data. Execute each query in MySQL Workbench and capture the actual result grids for submission.